

Chapter: 16

FIRST LAW OF THERMODYNAMICS**HEAT AND WORK:****HEAT FLOW: HEAT TRANSFER EQUATION**

In thermodynamics, thermal energy is associated with the internal energy of a system.

Heat (Q) is the transfer of thermal energy between systems due to a temperature difference. It flows spontaneously from a region of higher temperature to one of lower temperatures.

The rate at which heat is transferred can be quantified using the equation:

$$Q = mc\Delta T \dots\dots (1)$$

Where: Q = Heat transferred, m is the mass of the substance.
 c = Specific heat capacity of the substance
 ΔT = Temperature change.

The above equation, known as the heat transfer equation, highlights the dependence of heat transfer on the material properties and the extent of temperature change.

CONVERSION OF WORK TO THERMAL ENERGY:

Work done on a system can lead to an increase in its internal energy, contributing to thermal energy.

Explanation:

The mechanical work done generates heat due to friction, increasing thermal energy.

The concept of PV work can be illustrated by imagining a gas in a cylinder equipped with a movable piston as shown in figure.

The pressure exerted on the piston is essentially a force per unit area.

$$P = \frac{F}{A}$$

When the gas undergoes expansion, the piston moves upward, necessitating energy input (a force acting over a distance).

"The work done can be expressed as the product of force and distance".

In the case of a gas in a cylinder (thermal system), this simplifies the pressure multiplied by the change in volume.

$$W = P \cdot \Delta V \dots\dots (2)$$

Work done by the system is positive and work done on the system is negative.

$$W = - P \cdot \Delta V$$

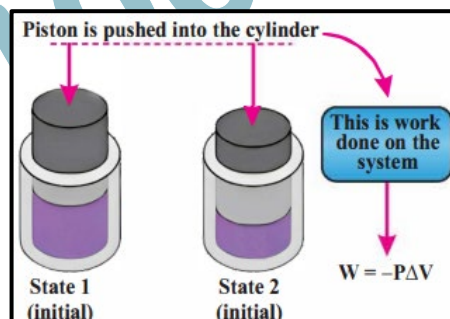
SELF-ASSESSMENT QUESTIONS:

1. Define heat is a form of energy.
2. Explain the basic concept of how work can be converted into thermal energy.

INTERNAL ENERGY:**Definition:**

"The sum of the random distribution of kinetic and potential energies within a system of molecules is called internal energy".

The symbol for internal energy is U and its SI unit is Joule (J).



Explanation:

Energy can generally be classified into two forms:

- 1) Kinetic Energy 2) Potential Energy.

The molecules of all substances contain both kinetic and potential energies.

"The amount of kinetic and potential energy of a substance depends on the phases of matter (solid, liquid or gas), this is known as the internal energy".

Determination of Internal Energy:

The internal energy of a system is determined by:

1. Temperature
2. The random motion of molecules
3. The phase of matter: gases have the highest internal energy; solids have the lowest
4. The internal energy of a system can increase by (1) Doing work on it (2) Adding heat to it
5. The internal energy of a system can decrease by losing heat to its surroundings

RELATION BETWEEN INTERNAL ENERGY & TEMPERATURE:

The internal energy of an object is intrinsically related to its temperature.

- When a gas in a container is heated, the gas molecules move faster, increasing their kinetic energy.
- In a solid, where molecules are tightly packed, heating causes the molecules to vibrate more.
- In both liquids and solids, molecules have both kinetic and potential energy due to intermolecular forces that keep them close together.

However, ideal gas molecules are assumed to have no intermolecular forces, meaning they only possess kinetic energy and no potential energy.

The change in internal energy of an ideal gas is equal to:

$$\Delta U = \frac{3}{2} n R \Delta T \quad (3)$$

Therefore, the change in internal energy is proportional to the change in temperature:

$$\Delta U \propto \Delta T$$

Where: ΔU = change in internal energy (J)

ΔT = change in temperature (K)

Conclusion:

"As the container is heated up, the gas molecules move faster with higher kinetic energy and therefore higher internal energy".

SELF-ASSESSMENT QUESTIONS:

1. How is the internal energy of an object related to its system?
2. How does an increase in temperature affect the internal energy of a substance?

THERMODYNAMICS:**THERMODYNAMIC:****Definition:**

"The study of the flow of heat or any other form of energy into or out of a system as it undergoes a physical or chemical transformation is called thermodynamics". OR

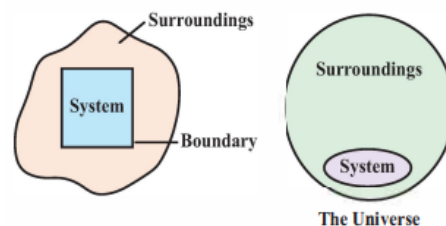
"The branch of physics which deals with the transformation of heat energy into other forms of energies is known as thermodynamics."

THERMODYNAMIC TERMS AND BASIC CONCEPTS:**THERMODYNAMIC SYSTEM:**

"A thermodynamic system is a specific portion of the physical universe that is chosen for analysis".

The system can be of any size or shape and can consist of a single substance, or a combination of substances as shown in the figure.

It is defined by boundaries and the surroundings.

**BOUNDARY:**

"The boundary is the physical or imaginary separation between the system and its surroundings".

SURROUNDING:

"The surroundings refer to everything outside the boundaries of the system".

This includes the rest of the universe that can interact with the system, separates the system from the rest of the universe.

TYPES OF THERMODYNAMIC SYSTEMS:**1. OPEN SYSTEM:**

"In an open system, both heat and matter can be exchanged with the surroundings".

For instance, steam (matter) can escape from the coffee, and heat can be lost to the surrounding air.

Example: A cup of hot coffee.

2. CLOSED SYSTEM:

"In a closed system, only heat can be exchanged with the surroundings, but matter cannot".

In this example, heat can still transfer through the pot, but the water vapor (matter) is trapped inside the pot and cannot escape.

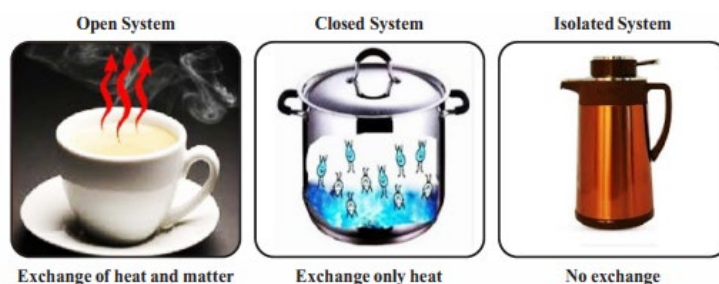
Example: A pot with a lid.

3. ISOLATED SYSTEM:

"In an isolated system, neither heat nor matter can be exchanged with the surroundings".

A thermos flask is designed to prevent heat transfer, keeping the contents inside hot or cold for a long period without any exchange with the environment.

Example: A thermos flask.

**WORK DONE BY GAS:**

When a gas expands, it does work on its surroundings by exerting pressure on the walls of the container it's in.

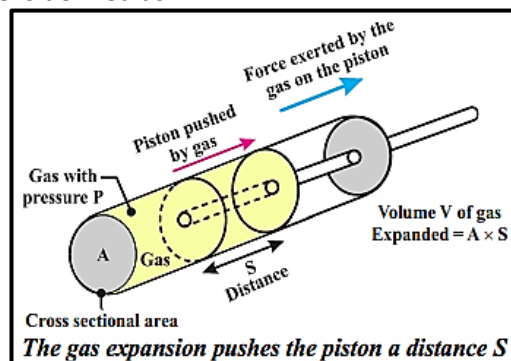
Explanation:

This is important (for example) in a steam engine, where expanding steam pushes a piston to turn the engine. The work done when a volume of gas changes at constant pressure is defined as:

$$W = P \cdot \Delta V$$

Where: ΔW = Work done (J)
 P = External pressure (Pa)
 ΔV = volume of gas (m^3)

For a gas inside a cylinder enclosed by a moveable piston, the force exerted by the gas pushes the piston outwards, Therefore, the gas does work on the piston



WORKED EXAMPLE: 16.2

When a balloon is inflated, its rubber walls push against the air around it. Calculate the work done when the balloon is blown up from 0.015 m^3 to 0.030 m^3 . Atmospheric pressure = $1.0 \times 10^5 \text{ Pa}$.

FIRST LAW OF THERMODYNAMICS:

Statement:

"In any thermodynamic process, when heat energy (ΔQ) is added to a system, this energy appears as an increase in the internal energy (ΔU) stored in the system plus the work done (ΔW) by the system on its surroundings".

Mathematical Representation:

Let " ΔQ " be the amount of heat energy added to a system, the energy will appear in the system as the increase in the internal energy " ΔU ", and the system will do some work " ΔW ". Then

$$\Delta Q = \Delta U + \Delta W$$

Where:

1. ΔQ is taken positive, when heat enters the system and negative, when it leaves the system.
2. ΔU is taken positive when the temperature of the system rises, while it is negative when the temperature of the system decreases.
3. ΔW is positive if work is done by the system and is negative if work is done on the system.

Explanation:

The first law of thermodynamics is based on the idea that:

"Energy can never be created nor be destroyed in any thermodynamic system, but it can be changed into other forms of energies, remaining the total amount of energies constant."

So, it is a particular form of the law of conservation of energy, which deals only with heat energy.

Since we know that energy appears in many forms such as mechanical energy, heat energy, sound energy, and electrical energy etc.

According to the law of conservation of energy, these energies may change from one form to another, without changing their total values.

EXPLANATION: BASED ON INTERNAL ENERGY:**Internal Energy:**

"The sum of all the kinetic (translational, rotational, and vibrational and potential energies associated with the random motion of the atoms of a substance or system is the internal energy (U) of the substance".

First law & Internal Energy:

According to the first law of thermodynamics:

"Every thermodynamic system possesses a state variable (U) called the internal energy".

The first law of thermodynamics states the change of internal energy, that:

"The change in internal energy of a system equals the net heat transfer into the system minus the net work done by the system".

In equation form, the first law of thermodynamics is:

$$\Delta U = \Delta Q - \Delta W \dots\dots\dots (1)$$

Where, ΔU = Change in internal energy of the system

ΔQ = Net heat transferred into the system

ΔW = Net work done by the system

It means that the change in the internal energy of a system is equal to the energy flowing in as heat minus the energy flowing out as work.

In other words, the change in the internal energy of the system is measured by the energy retained by the system.

The energy absorbed by the system changes the translational, vibrational, and rotational kinetic energy of the molecules. It also changes the potential energy of the molecules due to intermolecular forces.

Upon the initial and final states of the system and not on the path taken between these two states.

If the state of a thermodynamic system is changed from state (A) to state (B) then the change in internal energy of the system is.

$$\Delta U = U_B - U_A$$

Putting this value in equation (1), we have.

$$U_B - U_A = \Delta Q - \Delta W$$

APPLICATIONS OF THE FIRST LAW THERMODYNAMICS:**Principle:**

Applications of the first law of thermodynamics are all based on the principle that:

"The three forms of energy: internal energy; heat; and work can be interconverted".

The system is then often called the working substance.

The mathematical formulation of the law ($\Delta Q = \Delta U + \Delta W$) suggests the following processes that can be used for practical applications:

1. The processes in which any one of the three terms of the equation is zero.
2. The processes in which any one of the state variables P, V, T of the system are held constant.

1. ISOCHORIC PROCESS:**Definition:**

"The thermodynamics process during which the volume of the system remains constant is called the isochoric process".

Explanation:

Consider a gas contained in a cylinder having a conducting base and non-conducting walls and with a fixed piston at the end as shown in the Figure.

Let heat ΔQ is provided to the gas, and it heated at constant volume. The change in volume of gas is zero. i.e. $\Delta V = 0$.

The pressure of the gas increases from P_1 to P_2 while its temperature increases from T_1 to T_2 .

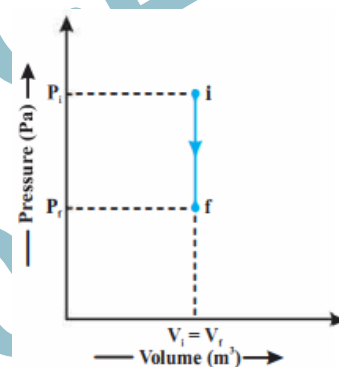
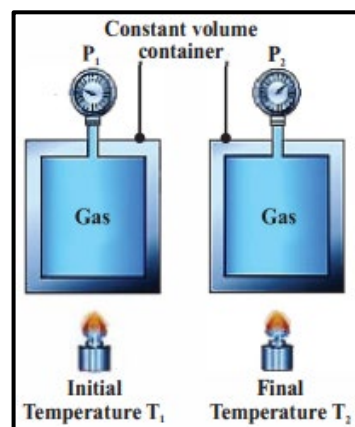
Since the piston is fixed, then the volume of the system will remain constant, and change of volume will be zero.

$$\begin{aligned} \text{As } \Delta W &= P \times \Delta V \\ &= P \times 0 \\ \Delta W &= 0 \end{aligned}$$

Since the system neither expands nor contracts, work is done neither by the system nor on the system i.e. $\Delta W = 0$.

Using the first law of thermodynamics equation.

$$\begin{aligned} \Delta Q &= \Delta U + \Delta W \\ \Delta Q &= \Delta U + 0 \\ \Delta Q &= \Delta U \end{aligned}$$

**Graphical Representation:**

The graph of isochoric process is called an "isochor", which is a straight line, parallel to the pressure axis.

Conclusion:

This means that in an isochoric process the entire amount of heat supplied to the gas is converted to the internal energy of the gas. The pressure and temperature of the gas will increase.

On the contrary, removal of heat from a system under isochoric conditions will cause an equivalent decrease in internal energy. The system will cool down and the pressure will fall.

2. ISOBARIC PROCESS:**Definition:**

"The thermodynamics process during which the pressure is kept constant is called an isobaric process".

Explanation:

Practically all heat engines depend on the transformation of heat into work.

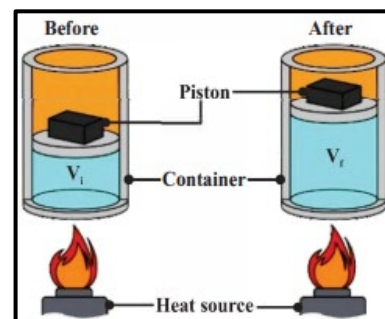
Isobaric expansion of the system is often used to convert heat into work.

Consider a gas contained in a cylinder having a conducting base and non-conducting walls and frictionless piston of cross-sectional area (A) as shown in figure.

When the gas is heated, a certain amount of heat energy is transferred into the system.

1. The gas expands and moves the piston outward.
2. The temperature changes from T_1 to T_2 . i.e., $\Delta T = T_2 - T_1$
3. Volume changes from V_1 to V_2 . i.e., $\Delta V = V_2 - V_1$

If the displacement of the piston is kept very small, the pressure of the gas will not change much and can be considered constant.



The work done by the gas which expands at constant pressure is:

$$\Delta W = P \Delta V$$

$$\Delta W = P (V_2 - V_1)$$

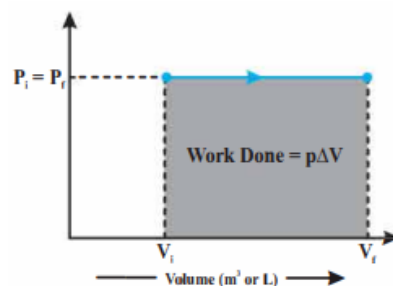
Using the first law of thermodynamics equation.

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = \Delta U + P \cdot \Delta V$$

Graphical Representation:

The graph of isobaric process is called an "isobar", which is a straight line, parallel to the volume axis.



Conclusion:

The work done by expanding or contracting gas comes from one or both sources:

1. Heat supplied to the gas and
2. The internal energy of the gas

3. ISOTHERMAL PROCESS:

Definition:

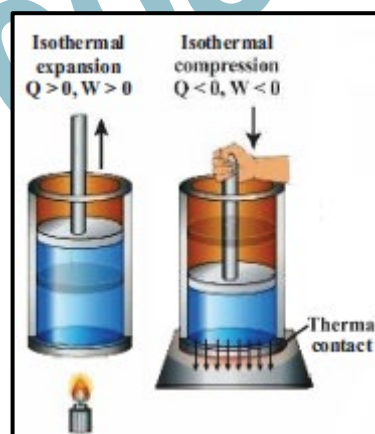
"The thermodynamics process which is carried out in such a way that a system undergoes changes, but its temperature remains constant is called an isothermal process".

Explanation:

The internal energy of an ideal gas does not depend on the volume but depends only on the temperature of the gas.

Consider a gas contained in a cylinder having a conducting base and non-conducting walls and with a movable piston at the end. The base of the cylinder is placed on a heat reservoir at temperature T_1 .

Let the gas be then allowed to expand slowly by decreasing the pressure on the piston. Due to this expansion, the gas tends to cool down. But heat is conducted from the heat reservoir to the gas so that the temperature of the system remains constant and is equal to the temperature of the reservoir.



Pressure changes from P_1 to P_2 , volume changes from V_1 to V_2 but temperature remains constant (T).

During isothermal expansion some work is done by the gas in pushing up the piston in the cylinder.

Since the temperature remains constant, there is no change in the internal energy of the gas, that is, $\Delta U = 0$

Using the first law of thermodynamics equation.

$$\Delta Q = \Delta U + \Delta W$$

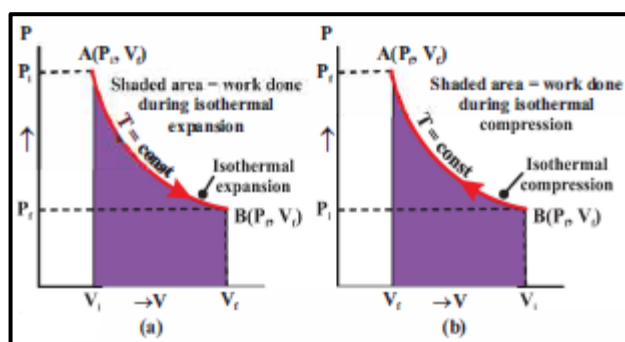
$$\Delta Q = 0 + \Delta W$$

$$\Delta Q = \Delta W$$

Graphical Representation:

In the PV diagram:

1. The work done during the isothermal expansion is equal to the area under the graph as shown in Figure.
2. Similarly, for an isothermal compression, the area under the PV graph is equal to the work done on the gas which turns out to be the area with a negative sign.



The whole process from an initial state to a final state is represented by a continuous curve which is called an "isotherm".

Conclusion:

This shows that:

1. If the gas expands and does external work, an equal amount of heat must be supplied in order to maintain its temperature constant.
2. Conversely, if the gas contracts, work is being done on it and an equal amount of heat must be allowed to leave the gas.

3. ADIABATIC PROCESS:

"The thermodynamics process during which no heat enters or leaves a system is called an adiabatic process".

For every adiabatic process $\Delta Q = 0$, a truly adiabatic process is an ideal one which cannot be realized.

However, the flow of heat may be prevented either by surrounding the system with a thick layer of heating insulating material such as cork, asbestos, or by performing the process very quickly.

The flow of heat requires finite time, so any process performed quickly enough will be practically adiabatic. To perform an adiabatic process on a gas, we consider the gas to be contained in a completely insulated cylinder with a movable piston at the end.

(a) Adiabatic Expansion:

If the gas is allowed to expand suddenly decreasing the pressure on the piston, the gas will be cooled, so that its temperature will decrease. Therefore, the internal energy of the gas will decrease.

In this case no heat is absorbed or given out by the system as it is insulated from the surroundings i.e. $\Delta Q = 0$.

And ΔW is positive, then according to the first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

$$0 = \Delta U + \Delta W$$

$$\Delta U = -\Delta W$$

The above equation can be written as. $\Delta W = -\Delta U$

This equation shows that internal energy (i.e., temperature) of the gas will decrease during expansion.

(b) Adiabatic Compression:

If the gas is compressed by suddenly increasing pressure on the piston, the gas will be heated, so that the temperature will increase. Therefore, the internal energy of the gas will increase and ΔW is negative.

According to the first law of thermodynamics.

$$\Delta Q = \Delta U + (-\Delta W)$$

$$0 = \Delta U - \Delta W$$

$$\Delta U = \Delta W$$

Thus, an increase in the internal energy of the system in an adiabatic process is equal to the work done on the system.

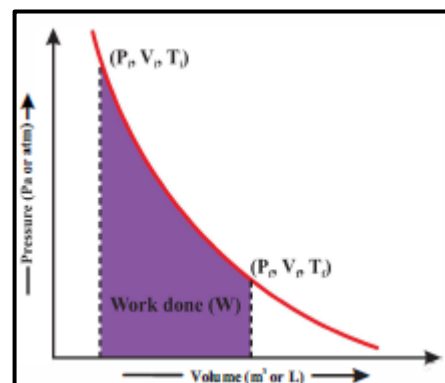
The above equation can be written as.

$$\Delta W = \Delta U$$

This equation shows that internal energy (i.e., temperature) of the gas will increase during compression.

Graphical Representation:

An adiabatic expansion is represented by the shaded curve in figure. The curve is known as an adiabat.



SELF-ASSESSMENT QUESTIONS:

1. What is the first law of thermodynamics and how is it expressed mathematically?
2. How does the first law of thermodynamics relate to the conservation of energy?

WORKED EXAMPLE: 16.3

A gas confined in a cylinder undergoes an expansion from an initial volume of 2.0 L to a final volume of 4.0 L against a constant external pressure of 2.0 atm. Calculate the work done by the gas during this process.

SELF-ASSESSMENT QUESTIONS:

1. How does an adiabatic process differ from an isothermal process in thermodynamics?
2. Why is an isothermal process important in thermodynamics, and how does it affect the internal energy of a system?
3. Why is an isobaric process significant in thermodynamics, and how does it affect the volume of a gas?

HEAT CAPACITY:

SPECIFIC HEAT:**Definition:**

"The measure of the heat energy required to raise the temperature of a unit mass (usually one gram or one kilogram) of a substance by one degree Celsius (or one Kelvin) is called specific heat".

Mathematically:

Specific heat, often denoted by C . The formula for specific heat (c) is given by:

$$Q = mc\Delta T$$

Where: Q = Heat energy absorbed or released,

m = Mass of the substance,

c = Specific heat, and

ΔT = Change in temperature.

Specific heat is an intensive property, meaning it does not depend on the quantity of the substance.

MOLAR SPECIFIC HEAT:**Definition:**

"The amount of heat energy needed to raise the temperature of one mole of a substance by one degree Celsius or Kelvin is called Molar specific heat". OR

Molar specific heat refers to the amount of heat energy required to change the temperature of 1 mole of a substance by 1 unit.

It is influenced by the substance's properties such as its nature, size, and composition.

Explanation:

When heat is absorbed by a body, its temperature increases, and when heat is lost, its temperature decreases. Since, temperature represents the total kinetic energy of the particles comprising the body. Therefore, when heat is absorbed, it increases the kinetic energy of these particles, thereby raising the temperature.

Mathematically:

Molar specific heat, denoted as C_m . It's calculated using the formula:

$$Q = nC_m\Delta T$$

Where: Q = Heat transferred,

n = Number of moles,

C_m = Molar specific heat,

ΔT = Temperature change.

Molar specific heat is an intensive property and relates to specific heat C through the equation:

$$C_m = MC$$

Where, M is the molar mass of gas.

TYPES OF MOLAR SPECIFIC HEAT OF GASSES:

There are two types of molar heat capacity:

1. C_p (molar heat capacity at constant pressure)
2. C_v (molar heat capacity at constant volume)

1. SPECIFIC HEAT AT CONSTANT VOLUME:

Definition:

"The amount of heat, required to raise the temperature of one mole of gas through 1 °C or 1K at constant pressure, is called specific heat of gas at constant pressure."

It is denoted by C_v .

$$C_v = \frac{\Delta Q_v}{n \cdot \Delta T}$$

2. SPECIFIC HEAT AT CONSTANT PRESSURE:

Definition:

"The amount of heat, required to raise the temperature of one mole of gas through 1 °C or 1K at constant volume, is called specific heat of gas at constant volume."

It is denoted by C_p .

$$C_p = \frac{\Delta Q_p}{n \cdot \Delta T}$$

RELATIONSHIP BETWEEN C_p AND C_v :

At constant pressure, the amount of heat supplied (Q_p) equal to the change in enthalpy (ΔH), expressed as:

$$Q_p = n C_p \Delta T$$

$$\Delta H = n C_p \Delta T \dots\dots\dots (1)$$

At constant volume, the amount of heat supplied (Q_v) equal to the change in internal energy (ΔU), expressed as:

$$Q_v = n C_v \Delta T$$

$$\Delta U = n C_v \Delta T \dots\dots\dots (2)$$

For one mole of an ideal gas ($n= 1$). Amount of heat supplied at constant pressure (Enthalpy) is

$$Q_p = n C_p \Delta T$$

$$\Delta H = (1) C_p \Delta T$$

$$\Delta H = C_p \Delta T \dots\dots\dots (3)$$

For one mole of an ideal gas ($n= 1$). The amount of heat supplied at constant Volume (Change in Internal energy) is

$$Q_v = n C_v \Delta T$$

$$\Delta U = (1) C_v \Delta T$$

$$\Delta U = C_v \Delta T \dots\dots\dots (4)$$

Using the first law of thermodynamics equation.

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta H = \Delta U + \Delta(PV)$$

$$= \Delta U + \Delta(RT)$$

Therefore, we have: $\Delta H = \Delta U + R \Delta T$ (5)

Substituting the equations (3) and (4) in equation (5), we get

$$C_P \Delta T = C_V \Delta T + R \Delta T$$

$$C_P = C_V + R$$

$$C_P - C_V = R$$

This equation shows the relationship between the molar heat capacities at constant pressure C_P and constant volume C_V for an ideal gas, where R is the universal gas constant.

WORKED EXAMPLE: 16.4

Calculate the molar specific heat capacity of a gas when 2 moles of the gas absorb 1500 J of heat energy, and its temperature increases by 25 degrees Celsius.

WORKED EXAMPLE: 16.5

A sample of water with a mass of 200 grams is heated, and its temperature rises from 25°C to 45°C. Calculate the heat transferred to the water. Given the specific heat capacity of water (c) is 4.18 J/g °C.

SELF-ASSESSMENT QUESTIONS:

- How does heat capacity relate to the amount of energy required to change the temperature of a substance?
- Explain why different materials have different heat capacities.
- How does the heat capacity of a substance change with temperature?

ASSIGNMENT: 16

MULTIPLE CHOICE QUESTIONS

SPECIFIC HEAT:

- The quantity of heat required to raise the temperature of unit mass of a substance through one degree centigrade is called: (2021)
(a) Molar specific heat (b) heat capacity (c) **Specific heat** (d) Latent heat
- The ratio of molar specific heat of a gas at constant pressure and molar specific heat at constant volume is: (2017)
(a) equal to one (b) less than one (c) **greater than one** (d) equal to zero
- The molar heat capacities of polyatomic gases as compared to the monatomic gases are: (2009)
(a) **Greater** (b) Smaller (c) Equal (d) Infinite
- The difference of molar specific heats at constant pressure and at constant volume per mole is called: (2005)
(a) Molar heat (b) Heat constant (c) **Boltzmann constant** (d) Gas constant
- The quantity of heat required for per degree rise in temperature of a body is called (2002M)
(a) Specific heat capacity (b) Heat of vaporization (c) Heat of fusion (d) **Heat capacity**

FIRST LAW OF THERMODYNAMICS:

- A thermodynamic process, such as compression or expansion done very quickly, will nearly be: (2022)
(a) Isothermal (b) Adiabatic (c) Isochoric (d) Isobaric
- A thermodynamic process in which heat exchange is not allowed is called: (2021)

- (a) Isochoric (b) Isobaric (c) Isothermal (d) Adiabatic
8. In this process no heat enters or leaves the system: (2014)
(a) Isochoric (b) Isobaric (c) adiabatic (d) Isothermal
9. The maximum work done is possible in this process: (2012, 06)
(a) Isochoric (b) Isochoric (c) Isobaric (d) Adiabatic
10. In Adiabatic expansion, the internal energy of the gas: (2011)
(a) remains the same (b) decrease (c) becomes zero (d) Increase
11. If no heat flows into or out of a system, the process is called: (2009)
(a) Isobaric (b) Isothermal (c) Isochoric (d) Adiabatic
12. Boyle's Law holds good for an ideal gas in a process called: (2009)
(a) Isobaric (b) Isochoric (c) Isothermal (d) Adiabatic
13. A thermodynamic process in which the change in volume of the system is zero tells that: (2008)
(a) the work done by the system is maximum (b) the work done on and by the system is zero
(c) the work done on the system is maximum. (d) none of the above.
14. A domestic pressure cooker is based on: (2005)
(a) Adiabatic process (b) Isothermal process (c) Isobaric process (d) Isochoric process
15. If the volume of the system remains constant during a process, it is called: (2004)
(a) Isochoric (b) Isothermal (c) Isobaric (d) Adiabatic
16. During an Adiabatic change, the pressure and the volume formula of a gas is given by: (2004)
(a) $PV = \text{const.}$ (b) $P^\gamma V = \text{const.}$ (c) $(PV)^\gamma = \text{const.}$ (d) $PV^\gamma = \text{const.}$
17. The internal energy of a system depends on; (2002E)
(a) Pressure (b) Volume (c) Temperature (d) Entropy
18. The process during which no external work is performed is: (2002E)
(a) Isothermal (b) Isochoric (c) Isobaric (d) Adiabatic
19. In an Isobaric Process the work done is equal to: (2002M)
(a) $\frac{Cv}{n\Delta T}$ (b) $\frac{Cp}{n\Delta T}$ (c) $\frac{R\Delta T}{n}$ (d) $nR\Delta T$
20. In an isothermal process, the internal energy of the system: (2001)
(a) Remains the same (b) Increases (c) Decreases (d) None of them

SUBJECTIVE QUESTIONS

FIRST LAW OF THERMODYNAMICS:

- State the first law of thermodynamics and explain it on the basis (i) Isobaric process (ii) Isothermal process.
- Briefly describe any two processes of thermodynamics. Draw the graph and give the mathematical equation for each.

SPECIFIC HEAT CAPACITY:

- Define heat capacity, specific Heat and Molar Specific Heat. Write down the mathematical equations and their units. Establish the relation between molar specific heat and common specific heat.
- What is molar Specific Heat? Why does a gas have two molar specific heats? Define them.
- Employing the 1st Law of Thermodynamics in two isotherms at different temperatures obtained in PV diagram show that $C_p - C_v = R$. OR

Using the first law of Thermodynamics, show that the sum of molar specific heat of a gas at constant volume and the molar gas constant is equal to the molar specific heat of the gas at constant pressure.

PROBLEMS

1. A system absorbs 1147 joules of heat, loses 233 joules of heat by conduction to the surroundings and delivers 614 joules of work. Calculate the change in the internal energy of the system. **(2018)**
2. In an Isobaric process 2000 J of heat energy is supplied to gas in a cylinder, the piston of area $2.0 \times 10^{-2} \text{ m}^2$ moves through 40 cm under a pressure of $1.01 \times 10^5 \text{ N / m}^2$. Calculate the work done and the increase in internal energy of the system. **(Ans: 1192 J) (2009, 02M)**
3. 1200 Joules of heat energy are supplied to the system at constant pressure. There internal energy of the system is increased by 750 joules and the volume by 4.5 cubic meters; find the work done against piston and the pressure on the piston. **(Ans: 450 J, 100 N/m²) (2003E, 94)**
4. If one mole of a mono-atomic gas is heated at constant pressure from -30°C to 20°C , find the change in its internal energy and the work done during the process. $C_p = 20.8 \text{ J/mole-k}$, $C_v = 12.5 \text{ J/mole}$
(Ans: 625 J, 415 J) (2001)
5. In an isobaric process, when 2000J of heat energy is supplied to a gas in a cylinder, the piston moves through 0.1m under a constant pressure of $1.01 \times 10^5 \text{ N/m}^2$. If the area of the piston is $5 \times 10^{-2} \text{ m}^2$, calculate the work done and the increase in the internal energy of the system. **(Ans: 1010 J, 990 J) (2000, 96)**

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